

HS4002: Lecture 2A

Crafting Research Questions

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Learning how to Submit Labs through Github

Why GitHub for Labs?

- Starting this week, you'll submit your weekly lab notebook through GitHub using the **fork-and-pull-request** workflow.
- It's the same workflow used by essentially every open-source project and most software teams — you're learning something you'll use again.
- Full reference for everything in this section: the `README.md` in the class repo (linked on NTULearn).

The Mental Model

- **Upstream** is my copy of the repo — `https://github.com/xiangyum/hs4002labs\_2026`. It's **public**, so anyone can view it, but you cannot push to it directly.
- **Your fork** is your own copy, living under your own GitHub account. You have full write access to it.
- A **pull request (PR)** is how you turn in the week's work — a link to your branch's diff against `main`.
- Grading happens *off* GitHub, in a spreadsheet — I won't be leaving PR comments. If something's wrong, I'll follow up with you directly.

One-Time Setup (Before This Week's Lab)

1. **Make a GitHub account** at <https://github.com> if you don't have one. Use a professional username — your name, not a nickname. This account will follow you into internships and jobs.
2. **Fork the repo:** go to https://github.com/xiangyum/hs4002labs_2026 and click **Fork**.
3. **Clone your fork**, then add the class repo as upstream.
4. **Authenticate** with `gh auth login` so you can push.

Worked Example: Resubmitting Week 1

- Let's walk through the weekly cycle using the Week 1 notebook as an example.
- Goal: make a small change, then submit it as a pull request, start to finish.

Step 1: Sync with Upstream

```
git checkout main  
git fetch upstream  
git merge upstream/main  
git push origin main
```

Makes sure your fork has the latest template before you start — do this every week, first.

Step 2: Create a Branch

```
git checkout -b week1
```

- Don't work directly on `main` — branching keeps each week's submission isolated and makes the PR diff clean.
- Switching branches can make files appear or disappear in Finder/Explorer — that's expected. Nothing is lost; it's just not the branch you're looking at.

Step 3: Make a Small Change

- Open `week1/week1_notebook.ipynb`.
- Add a markdown cell with a short note — e.g. one sentence on what you found interesting this week.
- Save the notebook.

Step 4: Commit and Push

```
git add week1/week1_notebook.ipynb  
git commit -m "Add reflection note to week 1 notebook"  
git push origin week1
```

Pushes your branch to *your fork* (origin) — not upstream. You can't push there.

Step 5: Open the Pull Request

- On GitHub, go to your fork — a banner offers “Compare & pull request” for the branch you just pushed. Click it.
- Confirm:
 - **base repository:** the class repo, base `main`
 - **head repository:** your fork, compare your branch
- Fill in the PR template (name, matric number, week) and submit.

That's It — You've Submitted

- The PR link *is* your submission — not a file you email me.
- No PR comments to expect — grading happens off GitHub, in the spreadsheet.
- Stuck on git itself (not the stats)? Ask — full troubleshooting is in README.md, and office hours exist for exactly this.
 - AI chatbots are also amazing at troubleshoot this.

What is a Research Question?

What is a Research Question?

“Questions about one or more topics or concepts that can be answered through research.” (Adler & Clark 2014)

What is a Research Question?

- Generally take the form of 5W1H.
 - But we might also be able to infer a research question from a thesis statement.
- They can be nested.
- What to pay attention to:
 - What are the core concepts?
 - What are the relations among the concepts?
 - What is the “level of analysis”?
 - Micro-level (person)
 - Meso-level (small group or organizations)
 - Macro-level (large groups or organizations)
 - Who is the intended audience?

Why Are They Important?

- As a reader:
 - Provides a frame for the content: it guides how you read, what you pay attention to.
 - It's a criteria of evaluation: am I interested in the paper's premise, is the paper delivering what it promises?
- As a researcher:
 - It's the premise of inquiry!

Classroom Exercise:

Figure 1: Kamal's *Representing Islam*



Identify as many research questions from this blurb of Prof. Nasir's book as you can. Describe the concepts involved, the relations among them, the level of analyses, and the intended audience of the questions.

How do Muslims who grew up after September 11 balance their love for hip-hop with their devotion to Islam? How do they live the piety and modesty called for by their faith while celebrating an art form defined, in part, by overt sexuality, violence, and profanity?

In *Representing Islam*, Kamaludeen Mohamed Nasir explores the tension between Islam and the global popularity of hip-hop, including attempts by the hip-hop ummah, or community, to draw from the struggles of African Americans in order to articulate the human rights abuses Muslims face. Nasir explores state management of hip-hop culture and how Muslim hip-hoppers are attempting to "Islamize" the genre's performance and jargon to bring the music more in line with religious requirements, which are perhaps even more fraught for female artists who struggle with who has the right to speak for Muslim women. Nasir also investigates the vibrant underground hip-hop culture that exists online. For fans living in conservative countries, social media offers an opportunity to explore and discuss hip-hop when more traditional avenues have been closed.



What I noticed

1. How do Muslims who grew up after September 11 balance their love for hip-hop with their devotion to Islam?
2. How do they live the piety and modesty called for by their faith while celebrating hip-hop?
3. What is the relationship between Islam and the global popularity of hip-hop?
4. How do the state and Muslim hip-hoppers "Islamize" hip-hop?
5. What does online underground hip-hop culture look like?

What Makes a Research Question Good?

Desirable Traits for RQs

- Clarity
 - Concepts, and the relations between them, are clearly identified.
- Tractability
 - It's clear what kinds of data would be needed to answer the question.
 - You are able to identify and use the methods needed for the analysis.
- Well-motivated
 - Why is this question worth asking?
 - Why is this question interesting?

- Known social problems or research puzzles.
- Problems you (or others around you) have encountered.
- Anything “puzzling” or unexpected you’ve encountered.

Pragmatic Tips For Improving Research Questions

Pragmatic Tips

1. Research is processual.
 - You refine RQs over time.
 - They usually become narrower; they may even pivot entirely.
2. Practice the art of “casing.”
 - What is ___ a case of?
 - This makes you connect your empirical interests with more general, but abstract, concepts.
3. Make the literature review work for you.
 - Identify empirical and theoretical “puzzles.”
 - Get an intuition for what your audience finds “interesting.”

How do you perform literature reviews? Break down your personal process step-by-step, and compare your workflow with those around you.



Literature Review Tips

1. Recognize quality sources of information.
 - Your literature review reflects your ability to curate a trustworthy source of information.
2. Read purposefully.
 - Always read with your research question in mind, or you can spend a life-time reading.
3. Start small and high quality.
 - Learn the principles of backward and forward search.
4. Use LLMs very cautiously.
 - LLMs can be extremely hit-or-miss because of copyright restrictions.

Thanks for listening :)